EXPERIMENT - 8

AIM :

To write a Python program to implement Depth-First Search (DFS) to traverse a graph by exploring one branch as deeply as possible before backtracking.

ALGORITHM:

1. Start.
2. Initialize a stack and a visited list.
3. Push the starting node onto the stack.
4. Pop a node from the stack.
5. If the node is not visited, mark it as visited and display it.
6. Push its unvisited adjacent nodes onto the stack.
7. Repeat until the stack becomes empty.
8. Stop.

PROGRAM :

untitled (3.14.3)

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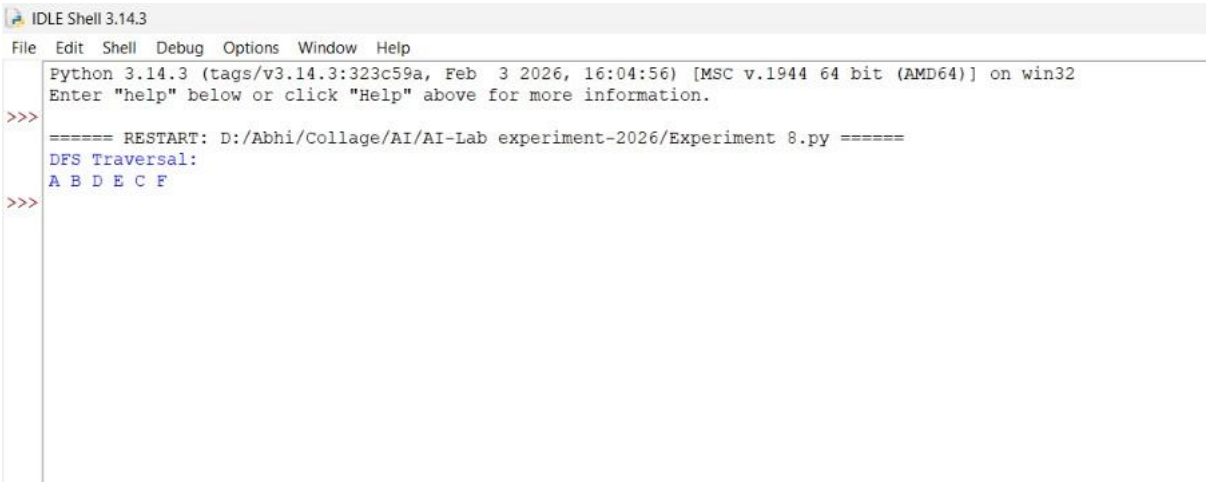
```
def dfs(graph, node, visited):
    if node not in visited:
        print(node, end=" ")
        visited.add(node)

        for neighbor in graph[node]:
            dfs(graph, neighbor, visited)

graph = {
    'A': ['B', 'C'],
    'B': ['D', 'E'],
    'C': ['F'],
    'D': [],
    'E': [],
    'F': []
}

print("DFS Traversal:")
dfs(graph, 'A', set())
```

OUTPUT:



RESULT:

Thus, the Python program to implement DFS was executed successfully, and the graph nodes were traversed using depth-first exploration.